Adult Cardiac Arrest

AC-03 | Adult Cardiac

Quick flow

01 Confirm Cardiac Arrest [EMT / AEMT / Paramedic]

Unresponsive • absent or abnormal breathing • no definite pulse within 10 seconds • honor valid DNR/MOST and obvious-death criteria.

02 Start TEAM-Focused CPR [EMT / AEMT / Paramedic]

Compressions 100-120/min • depth 2-2.4 in • full recoil • minimal pauses • change compressor every 2 min • chest-compression fraction goal >80%.

03 Attach AED / Monitor [EMT / AEMT / Paramedic]

Analyze rhythm immediately. Resume CPR after every rhythm check or shock. Check a pulse only with an organized rhythm and limit the check to 10 seconds.

04 Shockable Rhythm? [EMT / AEMT / Paramedic]

VF/pulseless VT → open AC-09. Asystole/PEA → open AC-01. Repeat rhythm analysis every 2 minutes.

05 Access + Medication [AEMT / Paramedic]

One rapid IV attempt, then IO if unsuccessful or delayed. Epinephrine 1 mg IV/IO every 3-5 min; timing follows the rhythm pathway.

06 Airway + Capnography [EMT / AEMT / Paramedic]

Two-person BVM first. Do not interrupt compressions for airway placement. After an advanced airway, give 1 breath every 6 sec with continuous compressions and waveform capnography.

07 Treat Reversible Causes [EMT / AEMT / Paramedic]

Actively evaluate and treat the Hs and Ts. Use cause-specific therapy rather than routine empiric arrest medications.

08 ROSC or Termination [EMT / AEMT / Paramedic / Medical Control]

ROSC → open AC-10. No ROSC → continue rhythm pathway and evaluate AC-12 when appropriate. Favor on-scene resuscitation unless a defined special circumstance supports transport.

Overview

- Master adult cardiac-arrest pathway for patients 16 years of age and older. Use this protocol to begin coordinated resuscitation and route immediately to the appropriate rhythm-specific pathway.
- Prioritize high-quality TEAM-focused CPR, rapid defibrillation when indicated, oxygenation and ventilation, guideline-directed medications, reversible causes, and on-scene resuscitation.

EMT Care

Arrest recognition, TEAM-focused CPR, AED care, two-person BVM ventilation, and basic reversible-cause treatment.

EMT

- Confirm unresponsiveness, absent or abnormal breathing, and no definite pulse within 10 seconds; identify valid DNR/MOST or obvious-death criteria.
- Activate additional resources and begin TEAM-focused CPR immediately, assuming or assigning compressor, airway, AED/monitor, recorder/timer, and team-lead functions.
- Perform compressions at 100-120/min to a depth of 2-2.4 inches with complete recoil, minimal interruptions, compressor changes every 2 minutes, and a chest-compression fraction goal above 80%.

- Apply the AED, deliver indicated shocks, and resume CPR immediately after every shock or rhythm analysis without an immediate pulse check.
- Provide two-person BVM ventilation with oxygen and an airway adjunct when available; maintain 30:2 until an advanced airway is in place.
- Assist with medication timing, documentation, reversible-cause treatment, family support, and safe movement only when indicated.

AEMT Care

All EMT care plus vascular access, epinephrine, supraglottic airway, and capnography when available.

AEMT

- Perform all EMT actions and make one rapid IV attempt without interrupting CPR or delaying defibrillation.
- If IV access is unsuccessful, infeasible, or delayed, establish IO access; use proximal tibia preferentially and humeral head only when tibial access is contraindicated.
- Give epinephrine 1 mg IV/IO every 3-5 minutes: as soon as possible for asystole/PEA and after the second shock for VF/pulseless VT.
- Place a supraglottic airway without interrupting compressions when BVM ventilation is ineffective or an advanced airway will improve care.
- After advanced-airway placement, ventilate once every 6 seconds during continuous compressions and use waveform capnography when available.

Paramedic Care

All AEMT care plus manual rhythm interpretation and defibrillation, definitive airway options, antiarrhythmics, and advanced reversible-cause treatment.

Paramedic

- Perform all EMT and AEMT actions and direct the rhythm-specific pathway, TEAM-focused resuscitation, scene strategy, and transition of care.
- For VF/pulseless VT, deliver an initial 200 J biphasic shock, verify the manufacturer-recommended setting, and use the same or higher energy for subsequent shocks; follow AC-09.
- After the third shock for persistent VF/pulseless VT, give either amiodarone or lidocaine according to AC-09; do not combine them.
- Choose a supraglottic airway or endotracheal intubation according to patient and scene conditions; airway placement must not interrupt compressions.
- Use continuous waveform capnography after any advanced airway and treat reversible causes with indication-specific therapy.
- Transition immediately to AC-10 after ROSC or apply AC-12 when termination criteria are met.

Indications

- Patient 16 years of age or older who is unresponsive, has absent or abnormal breathing, and has no definite pulse within 10 seconds.
- Adult patient in ventricular fibrillation, pulseless ventricular tachycardia, asystole, or pulseless electrical activity requiring resuscitation.

Contraindications

- Valid DNR, MOST, or other recognized order directing that resuscitation not be attempted.
- Obvious-death findings or circumstances addressed by agency policy.
- A definite pulse is present; use the appropriate perfusing-rhythm or medical protocol instead.

Assessment | Arrest Confirmation

- Assess responsiveness, breathing, and central pulse simultaneously; limit the pulse check to 10 seconds.
- If there is uncertainty about the pulse, begin CPR rather than delaying treatment.
- Identify the arrest as medical, traumatic, toxicologic, environmental, pregnancy-related, drowning-related, or associated with a mechanical circulatory device.

Assessment | CPR Quality

- Compression rate 100-120/min and depth 2-2.4 inches with complete recoil.
- Minimize pauses, keep rhythm and pulse checks under 10 seconds, and change compressors approximately every 2 minutes.
- Target a chest-compression fraction above 80% and use real-time CPR feedback when available.

Assessment | Rhythm Assessment

- Analyze rhythm as soon as the AED or monitor is available and approximately every 2 minutes thereafter.
- Route VF/pulseless VT to AC-09 and asystole/PEA to AC-01 using the direct protocol links.
- Check a pulse only when an organized rhythm is present or ROSC is otherwise suspected.

Assessment | Reversible Causes - Hs and Ts

- Hypovolemia, hypoxia, hydrogen ion excess/acidosis, hypo-/hyperkalemia, and hypothermia.
- Tension pneumothorax, cardiac tamponade, toxins, coronary thrombosis, and pulmonary thrombosis.

Treatment steps

- Begin high-quality CPR immediately and open AC-11 for TEAM-focused roles and performance targets.
- Apply AED/monitor pads and analyze the rhythm. Resume CPR immediately after every rhythm check or shock.
- For VF/pulseless VT, open AC-09 and prioritize rapid defibrillation. For asystole/PEA, open AC-01 and give epinephrine as soon as possible.
- For VF/pulseless VT, deliver an initial 200 J biphasic shock, verify the manufacturer-recommended setting, and use the same or higher energy for subsequent shocks according to the device.
- Perform rhythm analysis every 2 minutes. Check a pulse only with an organized rhythm or other evidence of ROSC, and limit the check to 10 seconds.
- Make one rapid IV attempt, then proceed to IO when IV access is unsuccessful, infeasible, or delayed. Proximal tibia is preferred; humeral head is an alternate only when tibial access is contraindicated.
- Give epinephrine 1 mg IV/IO every 3-5 minutes. Give it as soon as possible for nonshockable rhythms and after the second shock for VF/pulseless VT.
- For persistent VF/pulseless VT after the third shock, give either amiodarone or lidocaine according to AC-09. Do not combine the two agents.
- Provide two-person BVM ventilation first, maintaining 30:2 before an advanced airway and avoiding excessive ventilation.
- After advanced-airway placement, give 1 breath every 6 seconds during continuous compressions and use continuous waveform capnography.
- Use changes in waveform $EtCO_2$ as one component of assessment. An abrupt sustained rise may indicate ROSC; never use a single $EtCO_2$ value alone to terminate resuscitation.
- Identify and treat reversible causes. Give fluids, calcium, sodium bicarbonate, magnesium, antidotes, or other therapies only for a specific suspected or confirmed cause.
- Continue resuscitation on scene for most arrests. Transport during CPR only for a defined special circumstance or after sustained ROSC.

- Use mechanical CPR only when high-quality manual compressions are unsafe or impractical and device placement can occur with minimal interruption.
- If ROSC occurs, open AC-10. If resuscitation remains unsuccessful, open AC-12 and apply termination guidance when appropriate.

Medications

Epinephrine 1 mg/10 mL (0.1 mg/mL)

1 mg IV/IO every 3-5 minutes

- AEMT and Paramedic standing order for adult cardiac arrest.
- Give as soon as possible for asystole/PEA and after the second shock for VF/pulseless VT.

Amiodarone

300 mg IV/IO after the third shock for refractory VF/pulseless VT; may repeat 150 mg IV/IO once

- Paramedic standing order under AC-09.
- Choose amiodarone or lidocaine; do not combine them.

Lidocaine

1-1.5 mg/kg IV/IO after the third shock; then 0.5-0.75 mg/kg every 5-10 minutes; maximum total 3 mg/kg

- Paramedic alternative to amiodarone under AC-09.
- Choose lidocaine or amiodarone; do not combine them.

Warnings / precautions

- Do not interrupt CPR for IV/IO access, medication preparation, airway attempts, device placement, or prolonged rhythm checks.
- Do not transport routinely during active CPR. Movement commonly worsens compression quality and increases risk to the crew and public.
- Do not routinely administer calcium, sodium bicarbonate, magnesium, fluids, antidotes, or other cause-specific treatments without an appropriate indication.
- Naloxone must not delay CPR, defibrillation, airway support, or standard cardiac-arrest medications.
- Do not use routine head-up CPR, double-sequential defibrillation, vector-change defibrillation, or point-of-care ultrasound under this protocol.
- Do not use a single $EtCO_2$ cutoff or any single finding as the sole basis for terminating resuscitation.

Clinical pearls

- High-quality compressions and rapid defibrillation for a shockable rhythm provide the greatest immediate opportunity to improve outcome.
- Give epinephrine early for nonshockable arrest, but do not allow it to delay initial defibrillation attempts in VF/pulseless VT.
- A sudden sustained increase in $EtCO_2$ may signal ROSC; pause only for the scheduled brief rhythm and pulse check.
- Clear role assignment, closed-loop communication, visible medication and event timing, and structured debriefing improve team reliability.

Special populations | Pregnancy

- Begin standard resuscitation immediately. When the fundus is at or above the umbilicus, use continuous left uterine displacement, notify Medical Control early, and plan an obstetric-capable destination when clinically appropriate.

Special populations | Traumatic Arrest

- Use TB-10 because hemorrhage control, oxygenation, bilateral chest decompression, and transport decisions differ from primary medical arrest.

Special populations | Hypothermia, Drowning, or Toxicologic Arrest

- Use TE-05, TE-03, or TE-07 because pulse assessment, medication timing, resuscitation duration, antidotes, and transport decisions may differ.

Special populations | Mechanical Circulatory Device

- Use AC-14 or the applicable mechanical-circulation protocol; continuous-flow devices may not produce a palpable pulse, so assess perfusion and device function before starting compressions.

Related pathways / contacts

AC-09 VF / Pulseless VT - Open the shockable-rhythm pathway.

AC-01 Asystole / PEA - Open the nonshockable-rhythm pathway.

AC-11 TEAM-Focused CPR - Open team roles and CPR performance targets.

AC-10 Post-Resuscitation Care - Open immediately after ROSC.

AC-12 Termination of Resuscitation - Open when resuscitation remains unsuccessful.

AO-03 OB-GYN Emergency - Open for pregnancy-related causes, hemorrhage, seizure, or obstetric destination planning.

Source and references

American Heart Association. 2025 Adult Cardiac Arrest Algorithm. | American Heart Association. 2025 Guidelines for CPR and ECC: Adult Basic Life Support and Adult Advanced Life Support. | American Heart Association. 2025 Guidelines for CPR and ECC: Systems of Care. | American Heart Association. 2025 Guidelines for CPR and ECC: Adult and Pediatric Special Circumstances of Resuscitation. | Tennessee Emergency Medical Services Board. Tennessee EMS Protocol Guidelines, September 2025. | Claiborne EMS AC-01, AC-09, AC-10, AC-11, AC-12, medication formulary, and related special-circumstance protocols.

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